

STAT-437: APPLIED STATISTICAL METHODS

Assignment:1 solution

Total=40 Marks

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Question 1: A real-estate company wants to investigate factors that affect the selling price of a house. The company considers two explanatory variables:

$\mathbf{x}_1$ = house size (measured in hundreds of square feet);

$\mathbf{x}_2$ = number of bedrooms;

$\mathbf{y}$ = selling price (measured in thousands of dollars).

The company uses the multiple linear regression model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i, \quad i = 1, 2, \dots, n$$

where $E(\epsilon_i) = 0$ and $Var(\epsilon_i) = \sigma^2$.

In matrix form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where $\boldsymbol{\beta}^T = (\beta_0, \beta_1, \beta_2)$.

(i) Write the design matrix $\mathbf{X}$ for this multiple linear regression model.

Solution:

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{bmatrix} \rightarrow 2 \text{ marks}$$

(ii) The least-squares estimator of $\boldsymbol{\beta}$ is obtained by minimizing

$$S(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

Show that

$$S(\boldsymbol{\beta}) = \mathbf{y}^T \mathbf{y} - 2\boldsymbol{\beta}^T \mathbf{X}^T \mathbf{y} + \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}$$

Solution:

$$\begin{aligned} S(\boldsymbol{\beta}) &= (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \\ &= \mathbf{y}^T \mathbf{y} - \mathbf{y}^T \mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{y} + \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\beta} \\ &= \mathbf{y}^T \mathbf{y} - 2\boldsymbol{\beta}^T \mathbf{X}^T \mathbf{y} + \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\beta} \end{aligned}$$

where $\boldsymbol{\beta}^T \mathbf{X}^T \mathbf{y}$ is a 1×1 or a scalar and its transpose $(\boldsymbol{\beta}^T \mathbf{X}^T \mathbf{y})^T = \mathbf{y}^T \mathbf{X} \boldsymbol{\beta}$ is the same scalar.

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(iii) Differentiate $S(\beta)$ with respect to β and show that

$$\frac{\partial S}{\partial \beta} = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \beta$$

Solution: We know that $\mathbf{y}^\top \mathbf{X} \beta$ and $\beta^\top \mathbf{X}^\top \mathbf{y}$ are scalars and $\frac{\partial(\mathbf{X}^\top \beta)}{\partial \beta} = \frac{\partial(\beta^\top \mathbf{X})}{\partial \beta} = \mathbf{X}$

$$\begin{aligned} \frac{\partial(\beta^\top (\mathbf{X}^\top \mathbf{X}) \beta)}{\partial \beta} &= \frac{\partial}{\partial \beta} (\beta)^\top (\mathbf{X}^\top \mathbf{X}) \beta + \frac{\partial}{\partial \beta} (\beta)^\top [(\mathbf{X}^\top \mathbf{X})^\top \beta] \\ &= (\mathbf{X}^\top \mathbf{X}) \beta + [(\mathbf{X}^\top \mathbf{X})^\top \beta] \\ &= 2\mathbf{X}^\top \mathbf{X} \beta, \end{aligned}$$

→ 1 mark

where $\mathbf{X}^\top \mathbf{X}$ is symmetric. Hence

$$\begin{aligned} \frac{\partial S}{\partial \beta} &= 0 - 2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \beta \\ &= -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \beta \end{aligned}$$

→ 2 marks

(iv) By setting the derivative of $\frac{\partial S}{\partial \beta}$ equal to zero, show that the least-squares estimator is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Solution: By setting the derivative of $\frac{\partial S}{\partial \beta}$ equal to zero, we get

$$\begin{aligned} -2\mathbf{X}^\top \mathbf{y} + 2[\mathbf{X}^\top \mathbf{X}] \hat{\beta} &= 0 \\ \Rightarrow (\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{X}) \hat{\beta} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \\ \Rightarrow \hat{\beta} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \end{aligned}$$

→ 2 marks

where $\mathbf{X}^\top \mathbf{X}$ is invertible and $(\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{X}) = I$

(v) Show that $\hat{\beta}$ is an unbiased estimator of β .

Solution:

$$\begin{aligned} E(\hat{\beta}) &= E((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}) \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top E(\mathbf{y}) \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top E(\mathbf{X}\beta + \epsilon) \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}\beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top E(\epsilon) \\ &= \beta. \end{aligned}$$

→ 2 marks

(vi) Suppose the fitted regression equation is

$$\hat{y} = 40 + 2.5x_1 + 8x_2$$

Interpret the coefficients $\hat{\beta}_1 = 2.5$ and $\hat{\beta}_2 = 8$ in the context of the problem.

Solution:

Interpretation of $\hat{\beta}_1 = 2.5$: Holding the number of bedrooms constant, an increase of one unit in x_1 corresponds to an increase of 100 square feet in house size. Therefore, a 100-square-foot increase in house size is associated with an increase of 2,500 dollars in the predicted selling price.

→ 1 1/2 marks

Interpretation of $\hat{\beta}_2 = 8$: Holding house size constant, an increase of one bedroom is associated with an increase of 8,000 dollars in the predicted selling price.

→ 1 1/2 marks

(vii) Use the fitted model $\hat{y} = 40 + 2.5x_1 + 8x_2$ to predict the selling price of a house that has 2,000 square feet and 3 bedrooms.

Solution: The house has 2,000 square feet, but x_1 is measured in hundreds of square feet.

Therefore, $x_1 = \frac{2000}{100} = 20$ and $x_2 = 3$

Hence $\hat{y} = 40 + 2.5x_1 + 8x_2 = 40 + 2.5 \times 20 + 8 \times 3 = 114$ thousands of dollars.

→ 2 marks

Question 2: Teengamb Dataset

The dataset **teengamb**, available in the R package **faraway**, relates to a study of teenage gambling in Britain. Fit a multiple linear regression model with gambling expenditure as the response variable, and sex, status, and verbal score as the predictors.

(i) What is the residual mean square (MS_{Res}) for the fitted model?

```
rm(list=ls())
library(faraway)
library(foreign)
library(MASS)
library(olsrr)
data(teengamb)
# Response Variable
Y=teengamb[,5]
# Predictor Variables
sex=teengamb[,1]
status=teengamb[,2]
verbal=teengamb[,4]
#####
CC=cbind(sex,status,verbal)

# Form design matrix

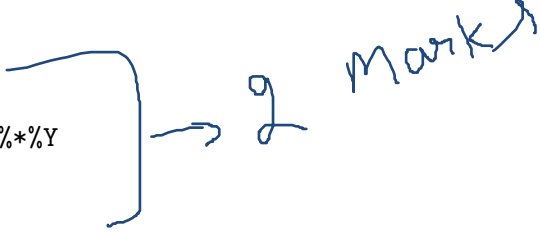
X=matrix(cbind(rep(1,length(sex)), CC), nrow=length(sex), ncol=4)

betahat=solve(t(X)%*%X)%*%t(X)%*%Y
betahat

# Fitting MLR

fit1=lm(Y~sex+status+verbal)
summary(fit1)

#####
###
SSRES= t(Y)%*%Y - t(betahat)%*%t(X)%*%Y
SSRES=33680.01
MSRES=SSRES/43
MSRES= 783.2559
```



(ii) Fit a multiple linear regression model to predict gambling expenditure among teenagers using sex, status, and verbal score as predictors.

```
> summary(fit1)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	69.2216	17.5619	3.942	0.000293	***
sex	-33.7520	9.6839	-3.485	0.001144	**
status	-0.4039	0.3267	-1.237	0.222971	
verbal	-2.7037	2.6784	-1.009	0.318410	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 27.99 on 43 degrees of freedom

Multiple R-squared: 0.2629, Adjusted R-squared: 0.2114

F-statistic: 5.111 on 3 and 43 DF, p-value: 0.004106

Hence the best fitted model for predicting gambling expenditure among teenagers using sex, status, and verbal score as predictors is $\text{gambling} = 69.2216 - 33.7520(\text{Sex}) - 0.4039(\text{status}) - 2.7037(\text{verbal})$.

(iii) Examine the presence of multicollinearity in the fitted model using the Variance Inflation Factor (VIF).

```
> ols_vif_tol(fit1)
```

Variables	Tolerance	VIF
sex	0.7378814	1.355231
status	0.5354771	1.867493
verbal	0.6886280	1.452163

The VIF of all predictors of teengamb data less than 4 that indicates no multicollinearity exist

(iv) What are the 95% confidence intervals for the coefficients of sex (β_1) and verbal score (β_3)?

A 95% confidence interval for the coefficient of sex (β_1) is

$$\begin{aligned}
 \hat{\beta}_1 \pm t_{43,0.025} SE(\hat{\beta}_1) &= \hat{\beta}_1 \pm qt(0.975, 43) SE(\hat{\beta}_1) \\
 &= -33.752 \pm 2.017 * 9.684 \\
 &= -33.752 \pm 19.533 \\
 &= (-53.285, -14.219).
 \end{aligned}$$

A 95 percentage confidence interval for the coefficient of verbal score (β_3) is

$$\begin{aligned} \hat{\beta}_3 \pm t_{43,0.025} SE(\hat{\beta}_3) &= \hat{\beta}_3 \pm qt(0.975, 43) SE(\hat{\beta}_3) \\ &= -2.704 \pm 2.017 * 2.678 \\ &= -2.704 \pm 5.402 \\ &= (-8.106, 2.698). \end{aligned}$$

(v) Does the Ordinary Least Squares (OLS) model provide a better fit to the data compared to the robust regression model?

```
fit1=lm(Y~sex+status+verbal)

summary(fit1)$sigma
[1] 27.987

fit3=rlm(Y~sex+status+verbal)

summary(fit3)$sigma
[1] 14.026
```

The residual standard error (RSE) for the Ordinary Least Squares (OLS) model is 27.987, whereas the RSE for the robust regression model is 14.026. Since a lower RSE indicates a better fit, the robust regression model provides a better fit to the data than the OLS model.

(vi) Identify the outliers using an Ordinary Least Squares (OLS) regression model with gambling expenditure as the response and sex and status as the predictors.

```
#####Finding outliers or influential observations###
> fit4=lm(Y~status+verbal)
> opar <- par(mfrow = c(2,2), oma = c(0, 0, 0.1, 0))
> plot(fit4, las = 1)
```

Figure 1 shows that observations 24, 33, and 36 are outliers to our model.

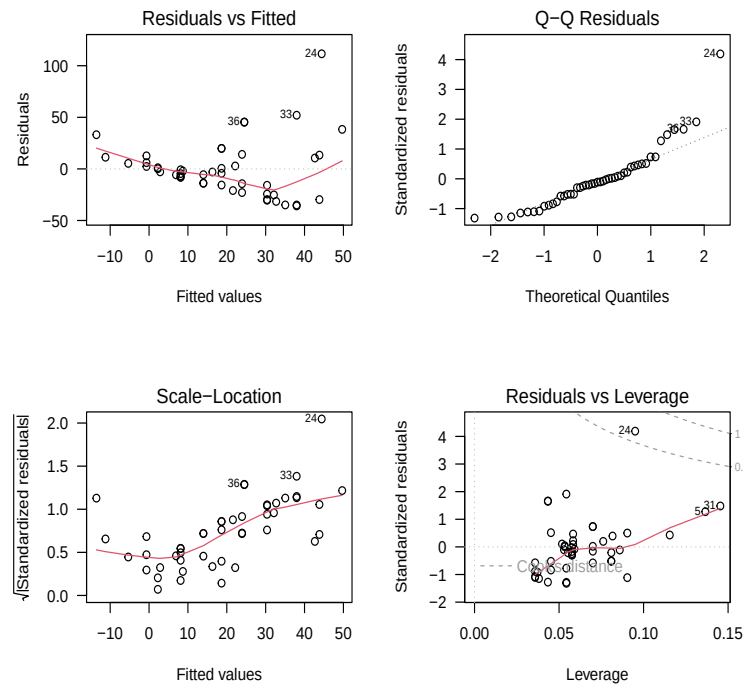


Figure 1: